

Electromagnetism 2

(spring semester 2026)

Lecture 2

Revision of vector calculus

- ❖ Gradient, divergence, curl and Laplace operators
- ❖ The del symbol
- ❖ Useful vector calculus theorems
- ❖ Laws of electrostatics and magnetostatics in the differential form

Previous lecture

- ❖ Laws of electrostatics and magnetostatics *in free space*, *in the integral form*, useful for practical computations:

Gauss law (*universally valid*)

$$\int_S \vec{E} d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho dV$$

Conservative nature of the **E**-field (*static field only*)

$$\oint_L \vec{E} d\vec{l} = 0$$

Absence of magnetic poles (*universally valid*)

$$\int_S \vec{B} d\vec{S} = 0$$

Ampere's law (*static field only*)

$$\oint_L \vec{B} d\vec{l} = \mu_0 \int_S \vec{j} d\vec{S}$$

Gradient of a scalar field

Gradient of a scalar field ($\text{grad}\varphi$) is a vector field:

- ✓ direction is perpendicular to equipotential surfaces;
- ✓ magnitude is equal to the derivative along that direction.

Along any unit vector $\vec{s}$, $\frac{\partial\varphi}{\partial s} = \vec{s} \cdot \text{grad}\varphi$

Equivalently, difference in field values at two nearby points **A** and **B**:

$$\delta\varphi = (\text{grad}\varphi) \cdot \delta\vec{r}$$

(scalar product of the gradient and the vector connecting **A** to **B**).

Therefore, the *slope is steepest* along the direction of gradient.

We know from calculus: $d\varphi = \frac{\partial\varphi}{\partial x}dx + \frac{\partial\varphi}{\partial y}dy + \frac{\partial\varphi}{\partial z}dz$

Therefore, in *any* Cartesian coordinates,

$$\text{grad}\varphi = \vec{e}_x \frac{\partial\varphi}{\partial x} + \vec{e}_y \frac{\partial\varphi}{\partial y} + \vec{e}_z \frac{\partial\varphi}{\partial z}$$

Divergence & curl of a vector field

Divergence of a vector field is a scalar field:

$$\operatorname{div} \vec{F} = \lim_{V \rightarrow 0} \left(\frac{1}{V} \int_{S(V)} \vec{F} d\vec{S} \right) \quad (\text{volume density of flux})$$

Curl of a vector field is a pseudovector field

(changes direction in a mirror image, defined in three dimensions only):

$$\operatorname{curl} \vec{F} = \lim_{V \rightarrow 0} \left(\frac{1}{V} \int_{S(V)} (d\vec{S} \times \vec{F}) \right) \quad (\text{volume density of overall circulation})$$

Equivalently, projection of the curl onto any unit vector $\vec{s}$:

$$(\operatorname{curl} \vec{F}) \cdot \vec{s} = \lim_{A \rightarrow 0} \left(\frac{1}{A} \oint_L \vec{F} d\vec{l} \right)$$

where the loop L lies in the plane orthogonal to $\vec{s}$

The del symbol

The **del** symbol (also known as **nabla**):

in *Cartesian coordinates*, $\nabla = \vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z}$

Gradient of a scalar field is a vector field:

$$\text{grad } \varphi = \nabla \varphi = \left(\frac{\partial \varphi}{\partial x} \vec{e}_x + \frac{\partial \varphi}{\partial y} \vec{e}_y + \frac{\partial \varphi}{\partial z} \vec{e}_z \right)$$

Divergence of a vector field is a scalar field:

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Curl of a vector field is a pseudovector field:

$$\begin{aligned} \text{curl } \vec{F} = \nabla \times \vec{F} &= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{e}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{e}_z \end{aligned}$$

The Laplace operator

Second derivatives of the fields: the *Laplace operator*

$$\nabla^2 = \left(\vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z} \right)^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

For a **scalar field**, the result is a **scalar field**:

$$\nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2}$$

For a **vector field**, the result is a **vector field**:

$$\begin{aligned} \nabla^2 \vec{F} &= \nabla^2 (\vec{e}_x F_x + \vec{e}_y F_y + \vec{e}_z F_z) = \\ &= \vec{e}_x \nabla^2 F_x + \vec{e}_y \nabla^2 F_y + \vec{e}_z \nabla^2 F_z \end{aligned}$$

Useful theorems (prove them!)

(T1) “div curl = 0”: $\nabla(\nabla \times \vec{F}) = 0$

(T2) “curl grad = 0”: $\nabla \times (\nabla \varphi) = \vec{0}$

(T3) “curl curl = grad div – del squared”

$$\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$$

Computation of divergence:

It is often convenient to use Cartesian coordinates:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Unit radial vector



For a radially/spherically symmetric vector field, $\vec{F} = F_r(r) \cdot \hat{r}$,

(T4) in cylindrical coordinates (r, φ, z) , $\nabla \cdot \vec{F} = \frac{1}{r} \frac{\partial(r F_r)}{\partial r}$

(T5) in spherical coordinates (r, φ, θ) , $\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r}$

Vector calculus theorems

Relations of physics inside a volume to physics on its surface,
and physics on a surface to physics on its boundary.

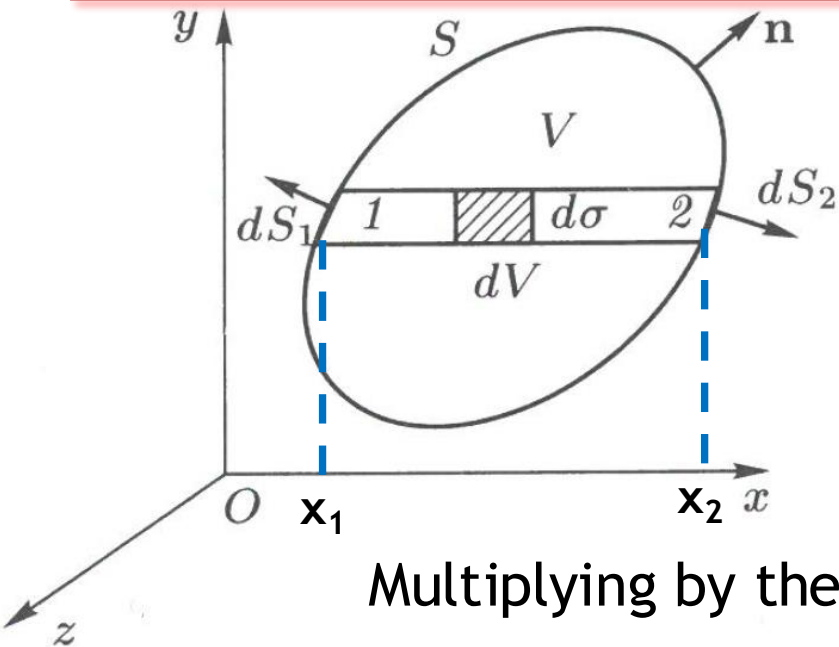
Divergence (Gauss) theorem:

$$\underbrace{\int_S \vec{F} d\vec{S}}_{\text{Flux of a vector field through a closed surface}} = \underbrace{\int_V (\nabla \cdot \vec{F}) dV}_{\text{Integral of its divergence over the volume enclosed}}$$

Curl (Kelvin–Stokes) theorem:

$$\underbrace{\oint_L \vec{F} d\vec{l}}_{\text{Line integral of a vector field around a closed curve}} = \underbrace{\int_S (\nabla \times \vec{F}) d\vec{S}}_{\text{Flux of its curl through any surface enclosed by the curve}}$$

Proof of divergence theorem (1)



[not discussed in the lecture]

For a scalar field $f(x, y, z)$,

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial x} dx = f_2 - f_1$$

Multiplying by the cross-section $d\sigma > 0$,

and using $d\sigma = -n_{1x}dS_1 = n_{2x}dS_2$,

where $\vec{n}$ is a unit normal vector to surface,

$$\int_{dV} \frac{\partial f}{\partial x} dV = (f_2 - f_1)d\sigma = f_1 n_{1x} dS_1 + f_2 n_{2x} dS_2 = \int_{dS_1 + dS_2} f n_x dS$$

Summing over elementary volumes, $\int_V \frac{\partial f}{\partial x} dV = \int_S f n_x dS$

Proof of divergence theorem (2)

[not discussed in the lecture]

For a vector field $\vec{F}(x, y, z)$,

$$\int_V \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right) dV = \int_S (F_x n_x + F_y n_y + F_z n_z) dS$$

$$\int_V \nabla \vec{F} dV = \int_S (\vec{F} \vec{n}) dS$$

Using the vector area $d\vec{S} = \vec{n} dS$,

$$\underbrace{\int_S \vec{F} d\vec{S}}_{\text{Flux of a vector field through a closed surface}} = \underbrace{\int_V (\nabla \cdot \vec{F}) dV}_{\text{Integral of its divergence over the volume enclosed}}$$

Flux of a vector field
through a closed surface

Integral of its divergence
over the volume enclosed

Differential laws for static fields

Gauss law and the divergence theorem for **any** volume **V**:

$$\int_S \vec{E} d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho dV = \int_V \nabla \vec{E} dV$$

Therefore $\nabla \vec{E} = \text{div} \vec{E} = \rho/\epsilon_0$; similarly $\nabla \vec{B} = \text{div} \vec{B} = 0$

Conservative nature of the electrostatic field **E**, and curl theorem for **any** loop **L** and **any** surface **S** bounded by the loop:

$$\oint_L \vec{E} d\vec{l} = \int_S (\nabla \times \vec{E}) d\vec{S} = 0$$

Therefore, for the electrostatic field, $\nabla \times \vec{E} = \text{curl} \vec{E} = 0$

Similarly, **Ampere's law** becomes $\nabla \times \vec{B} = \text{curl} \vec{B} = \mu_0 \vec{j}_{10}$

Field potentials

For the electrostatic field, $\nabla \times \vec{E} = 0$

For any scalar field (T2), $\nabla \times (\nabla \varphi) = 0$

Therefore, one can define a *scalar potential* of the electrostatic field (the minus sign is a convention):

$$\vec{E} = -\nabla \varphi$$

The electrostatic field is *irrotational* or *conservative*.

For the magnetic field, $\nabla \cdot \vec{B} = 0$ (the field is *solenoidal*)

For any vector field (T1), $\nabla(\nabla \times \vec{A}) = 0$

Therefore, one can define a *vector potential* of the magnetic field:

$$\vec{B} = \nabla \times \vec{A}$$

Definitions of the potentials are not unique (“gauge freedom”).

More details: *lecture 3*.

Summary

- ❖ Laws of electrostatics and magnetostatics *in free space, in the differential form*:

Gauss law (*universally valid*)

$$\nabla \vec{E} = \rho / \epsilon_0$$

Conservative nature of the **E**-field (*static field only*)

$$\nabla \times \vec{E} = 0$$

Absence of magnetic poles (*universally valid*)

$$\nabla \vec{B} = 0$$

Ampere's law (*static field only*)

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

- ❖ Electrostatic field is described by a scalar potential, $\vec{E} = -\nabla \varphi$
- ❖ Magnetic field is described by a vector potential, $\vec{B} = \nabla \times \vec{A}$